National Yang Ming Chiao Tung University
電子研究所 / 前瞻半導體研究所
Physical Design Autmation, Spring 2026

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Programming Assignment #3

Placement with OpenROAD

Due: May 25, 2026

1 Problem Description

In the IC design flow, floorplanning fixes the macros, while global placement determines the approximate cell positions. Crucially, global placement does not produce a legalized layout, as it prioritizes design objectives over overlap avoidance. These overlaps are often further complicated by subsequent optimizations, such as Multi-Bit Flip-Flop banking and Buffer Insertion. Therefore, legalization and detailed placement are essential to transform these inherently illegal distributions into a valid, rule-compliant layout.

Let $B = \{b_1, b_2, b_3, \ldots, b_n\}$ be a set of unmovable macros and blockages, and $C = \{C_1, C_2, C_3, \ldots, C_n\}$ be a set of movable cells. The objective of the legalization and detailed placement problem is to place all cells within a rectangular chip area such that they are non-overlapping and strictly aligned with the site rows. The placement must minimize the average displacement of cells while maintaining a uniform density distribution across the chip.

2 Input

The input file starts with DBU_per_micron, followed by the die area coordinates and site dimensions. After a mandatory blank line, a header line defines the column fields: Name LLX LLY Width Height Type. All subsequent lines provide the actual data for each instance, following the sequence specified by the header. The input format and a sample input are given as follows:

Input Format
DBU_Per_Micron # // Number hard IP blocks
DieArea_LL # // Lower-left coordinate of the die
DieArea_UR # // Upper-right coordinate of the die
Site_Width # // Width of each site
Site_Height # // Height of each site
// Empty line
Name LLX LLY Width Height Type
<instName> <lowerleftX> <lowerleftY> <cellWidth> <cellHeight> <cellType>
⋮
Sample Input
DBU_Per_Micron 1000
DieArea_LL 0 0
DieArea_UR 1500000 1500000
Site_Width 200
Site_Height 2000
Name LLX LLY Width Height Type
FE_OCPC15395_n_142781 1122957 200112 400 2000 CELL
FE_OCPC15396_n_142781 1296923 199847 400 2000 CELL
⋮
h0 50655 10370 755800 382000 MACRO
h1 55655 488000 775000 484000 MACRO
⋮
B0 -800 0 52000 1498000 BLOCKAGE
B1 1432800 0 66600 1498000 BLOCKAGE
⋮

3 Output

In the program output, you are asked to give the bottom-left coordinate of each cell in OpenROAD TCL format. The following table gives the output format and an output to the sample input.

Output Format
<place_cell> -inst_name <instName> -orient R0 -origin $\{X\ Y\}$ ⋮
Sample Output
place_cell -inst_name inst6050 -orient R0 -origin {49.4 35.8} place_cell -inst_name inst5821 -orient R0 -origin {51 36.8} place_cell -inst_name inst6286 -orient R0 -origin {42.9 37.6} ⋮

4 Language/Platform

1. Language: C or C++ is preferred.

2. Platform: Linux.

5 Command-line Parameter

In order to test your program, you are asked to add the following command-line parameters to your program: Legalizer <alpha> <threshold> <input_file> <output_file>. Note that the file names must be based on the design name (e.g., <designName>_insts.gp as input and <designName>_insts.tcl as output). The TA will run your program with:

```
$ make
$ ./Legalizer <alpha> <threshold> <input>.gp <output>.tcl
```

6 Submission

You need to submit a tar file, which includes (1) source codes and (2) Makefile to E3 (`https://e3.nycu.edu.tw/`) by the deadline. Please put all required files in a folder, and use the following command to compress the folder in the linux environment.

```
$ tar cvf Student_ID.tar Student_ID
```

The folder name and the compressed file name must be your student ID.

7 Grading Policy

This programming assignment will be graded based on (1) the **legality** and (2) **solution quality**. For each test case, it will be regarded as "failed" if your program takes more than 30 mins.

The final evaluation is conducted across four benchmarks: two public and two hidden cases. For each benchmark, the performance is assessed using two distinct parameter configurations: one emphasizing Total

Displacement and the other prioritizing Density Overflow Ratio (DOR). The Quality of a legalized placement is determined by a weighted cost function, where a lower value indicates a superior result. To quantify congestion, we calculate the DOR as the percentage of $10\mu m \times 10\mu m$ grids exceeding a density threshold T. Notably, regions occupied by fixed macros are excluded from the grid count to accurately reflect standard cell congestion. Each case's total score is a combination of a base score for legality and a performance-based score derived from the quality rankings across all eight test runs (4 benchmarks $\times$ 2 parameter sets).

$$\text{Total Score} = 70(\text{legality pass}) + \text{Performance Score}_{\text{rank}}$$

$$\text{Quality} = \alpha \cdot \text{Average Displacement} + (1-\alpha) \cdot \text{DOR, lower is better.}$$

$$\text{DOR} = \frac{\text{Number of Overflow Grids (Excluding Macros)}}{\text{Total Number of Grids (Excluding Macros)}} \cdot 100$$

There will be 20% penalty per day for late submission.

8 Notes

1. Direct execution of **detailed_placement** in your final output TCL script is strictly prohibited. TAs will verify your scripts; violations will result in a score of zero. However, you are encouraged to use this command for observation or debugging purposes during development.
2. Cell rotation is forbidden. All movable cells must maintain their original orientation during the process.
3. The input and output filenames must strictly follow the required format (e.g., `<designName>_insts.gp` and `<designName>_insts.tcl`); otherwise, a 5-point deduction will be applied for each case.
4. Please **DO NOT** use VSCode SSH to connect to the server; use MobaXterm instead. VSCode SSH prevents GUI display and places excessive overhead on the server.
5. Plagiarism is strictly forbidden.